MedSim Eval: speaker notes

Overview deck · 9 slides · about 13 minutes, leaving time for questions in a 15-minute slot

One page per slide: **talking points** to glance at, the **line into the next slide**, and a short **script** to fall back on. Key figures and likely questions are at the end.

#	Slide	Time	Ends at	The one thing to land
1	**MedSim Eval**	0:30	0:30	MedSim Eval tests any space-medicine assistant in simulated scenarios built on NASA data.
2	**The evaluation stack, from scenario to report**	1:30	2:00	Four layers; authoring is built, the rest is designed.
3	**How a scenario runs, from authoring to reports**	2:00	4:00	A scenario flows from authoring, through simulation, to scored reports, and every step is logged.
4	**Every scenario is grounded in NASA's own data**	1:30	5:30	Everything rests on NASA's own medical data; anything assumed is labelled.
5	**Each scenario runs up to 12 times, in 5 settings**	1:30	7:00	Each scenario is tested in every environment where it can happen: up to 12 runs.
6	**Four layers of scoring, reported side by side**	1:30	8:30	Four kinds of score, reported side by side, each backed by evidence.
7	**Worked example: appendicitis, 8 hours to act**	2:00	10:30	One scenario, end to end: timing decides the outcome.
8	**Plan of work: six phases, authoring to reports**	1:30	12:00	Six phases; authoring is built, and the runner and scoring come next.
9	**What we need to keep moving**	1:00	13:00	Four asks; the Scenario Builder is ready now.

MedSim Eval

MedSim Eval tests any space-medicine assistant in simulated scenarios built on NASA data.

TALKING POINTS

- MedSim Eval tests **any** space-medicine assistant (SpaceDoc, CMO-DA, others) in **simulated medical scenarios**.
- Built on NASA's own medical risk data: the **Integrated Medical Model**.
- Today: what's in the stack, how a scenario runs, a worked example, and the plan of work.

NEXT “Let's start with the stack and what's already built.”

SCRIPT

Thanks for having me. MedSim Eval is an evaluation system for space-medicine digital assistants: SpaceDoc, CMO-DA, or anything else. Instead of quizzing an assistant with fixed questions, we put it in simulated medical scenarios, built from NASA's own medical data, and score how it handles them.

I'll cover what's in the stack, how a scenario runs, a worked example, and the plan of work.

The evaluation stack, from scenario to report

Four layers; authoring is built, the rest is designed.

TALKING POINTS

- Four layers, top to bottom: **authoring**, **simulation**, **scoring**, **reporting**.
- **Built:** scenario authoring: the Scenario Builder clerks use, a validator, and a NASA-sourced library.
- **Designed, not yet built:** simulation (digital twin, AI crewmember, runner), scoring (scorer, judge, viewer), reporting.
- Right panel, three principles: works with **any assistant**; **runs locally**, on a laptop overnight, no cloud; **every score is traceable** to the transcript turn or twin event behind it.

NEXT “Here's how those pieces fit together when a scenario runs.”

SCRIPT

MedSim Eval is four layers, from a written scenario to a report. The first, scenario authoring, is built: clerks write scenarios in the Scenario Builder, a validator checks them, and everything they pick from comes from a NASA-sourced library.

The other three are designed but not built yet: simulation, where a digital twin of the patient and an AI crewmember play out the scenario; scoring, where every run is checked against a rubric; and reporting.

Three principles run through all of it. It works with any assistant, through a small adapter. It runs locally, on a laptop overnight, with no cloud calls. And every score is traceable to the exact moment in the conversation or the simulation behind it.

How a scenario runs, from authoring to reports

A scenario flows from authoring, through simulation, to scored reports, and every step is logged.

TALKING POINTS

- Top left, **authoring**: clerks write a scenario from the NASA library; the physiologist reviews flagged ones.
- Middle, **simulation**: an **AI crewmember** plays the patient or the CMO; the **digital twin** holds the patient's physiology, the onboard kit and a simulated clock.
- Right, the **assistant under test**: it talks to the crewmember, and can query the twin through **MCP tools** for vitals, exams, tests and treatments.
- Everything is **logged** with its simulated time.
- Bottom row, left to right: **scorer** (each item, with evidence), **judge** (all runs, flags anomalies), **clinician review** (viewer jumps to the exact turn), **feedback agent**, then the **medical** and **technical** reports.
- Every run has a seed, so any run can be **replayed exactly**.

NEXT “All of this rests on NASA's own data.”

SCRIPT

Here's a scenario running end to end. On the left, clerks write the scenario, drawing on the NASA library, and our physiologist reviews any that need it.

In the middle is the simulation. An AI plays the crewmember, either the patient or the crew medical officer, and a digital twin holds the patient's physiology, the medical kit aboard and a simulated clock. The assistant under test, on the right, has two ways in: it talks to the crewmember, and if it supports tools it can query the twin directly for vitals, exams, tests and treatments. Everything is logged with its simulated time.

Along the bottom, a scorer checks every rubric item in every run and has to cite the evidence; a judge reads all the runs and flags anything unusual; clinicians review those flags in a viewer that jumps straight to the moment in question; and a feedback agent writes two reports, one medical and one technical. Because every run has a seed, any run can be replayed exactly.

Every scenario is grounded in NASA's own data

Everything rests on NASA's own medical data; anything assumed is labelled.

TALKING POINTS

- **118 conditions** from NASA's Integrated Medical Model: the 2023 **CLiFFs**, clinical fact sheets with treated and untreated courses.
- **17** Design Reference Missions, **8** medical kit profiles, **20** synthetic crewmembers modelled on the active astronaut corps.
- **678** ICD-10-CM symptom codes and **90** reference entries (kit drugs from FDA labels, plus terms).
- Chart: how many of the 118 can occur in each environment: **100** in LEO, **89** cislunar and Mars transit, **108** on the lunar surface, **106** on Mars.
- Anything not from a NASA source is **labelled as an assumption**; course numbers are pending physiologist review.

NEXT “Clerks don't pick a mission. The system places each scenario in every environment where it can happen.”

SCRIPT

None of this is invented from scratch. The conditions come from NASA's Integrated Medical Model: 118 conditions, each with a clinical fact sheet, a CLiFF, that says how it progresses with and without treatment.

We also have 17 of NASA's Design Reference Missions, eight medical kit profiles for different vehicles and habitats, 20 synthetic crewmembers modelled on the active astronaut corps, the standard ICD-10 symptom codes, and reference entries for the kit drugs, taken from FDA labels.

The chart shows how many of the 118 conditions can occur in each environment, from 89 in cislunar space and Mars transit up to 108 on the lunar surface. Anything we had to assume is labelled as an assumption, and the course numbers are waiting on physiologist review.

Each scenario runs up to 12 times, in 5 settings

Each scenario is tested in every environment where it can happen: up to 12 runs.

TALKING POINTS

- Runs in **all five environments** where the condition can occur: LEO, cislunar, lunar surface, Mars transit, Mars surface.
- In each, the CLiFF's **best** and **worst** course, plus **2 runs at the CLiFF's own odds**: up to **12 runs** per scenario.
- Each environment brings its own **kit, gravity and communication delay**. Chart: ISS **84** drug and **142** equipment types; Orion **55 / 62**; lunar lander **35 / 23**.
- An item needing equipment that isn't aboard is **not applicable**, not failed.
- The **clock jumps ahead** when nothing is happening: a 10-day antibiotic course takes seconds.

NEXT “Each of those runs is then scored four ways.”

SCRIPT

Clerks don't choose a mission; the system does. Each scenario runs in every environment where its condition can occur, from low Earth orbit to the Mars surface.

In each one it runs twice, once on the CLiFF's best-case course and once on the worst, and then twice more at the CLiFF's own odds. That's up to 12 runs per scenario.

Each environment brings its own kit, gravity and communication delay, and the kits really differ: the ISS carries 84 drug types and 142 equipment types, the lunar lander 35 and 23. If a checklist item needs equipment that isn't aboard, it's marked not applicable rather than failed. And because the clock jumps ahead when nothing is happening, a ten-day antibiotic course runs in seconds.

Four layers of scoring, reported side by side

Four kinds of score, reported side by side, each backed by evidence.

TALKING POINTS

- **A, critical errors:** six never events (e.g. a drug the patient is allergic to). Any one **fails the run**.
- **B, checklist:** observable actions written by clerks, weighted **−10 to +10**.
- **C, global ratings:** six domains, each rated **1–5** against written descriptions.
- **D, outcome:** how the simulated patient did: course reached, time to diagnosis and treatment.
- Chart: the worked example's points sit mostly in **management** and **safety**; the biggest penalties are for **Earth-only plans** (surgery, CT, evacuation).
- Never one headline number. The judge **flags outlier runs** instead of averaging them away.

NEXT “Here's what that looks like on one scenario.”

SCRIPT

Every run is scored four ways, and the four are reported side by side rather than rolled into one number.

First, critical errors: six never events, like giving a drug the patient is allergic to. Any one of them fails the run. Second, a checklist of observable actions the clerk writes for the scenario, each weighted from minus ten to plus ten. Third, global ratings: six domains, from information gathering to communication, each rated one to five against written descriptions. Fourth, the outcome: how the simulated patient actually did.

The chart shows where the points sit in our worked example: most are in management and safety, and the largest penalties are for plans that only work on Earth, like surgery or evacuation. The scorer goes one item at a time and cites its evidence, and the judge flags runs that differ from the rest instead of averaging them away.

Worked example: appendicitis, 8 hours to act

One scenario, end to end: timing decides the outcome.

TALKING POINTS

- **Appendicitis** in a synthetic crewmember, **Kenji Arakawa, 45**, with a **penicillin allergy**.
- Reported by a **non-physician CMO** who leaves details out unless asked; the patient blames food.
- **T+0** periumbilical pain; **T+6h** pain moves to the right lower quadrant; **T+8h** deadline for antibiotics that cover gut bacteria.
- **Complication events:** untreated at **T+36h**, perforation with peritonitis; a **penicillin** given, an allergic reaction in some runs.
- Chart, CLiFF worst case: risk of loss of crew life up to **8.7%** if treated in time, up to **50%** if not.
- Scored on **19 checklist items** plus the standard items (diagnosis, cause, and each complication).

NEXT “So that's the system. Here's the plan to build the rest.”

SCRIPT

Here's the worked example. The patient is Kenji Arakawa, a synthetic 45-year-old crewmember with appendicitis and a penicillin allergy. The assistant hears about it from a non-physician CMO who leaves details out unless asked, and the patient blames something eaten.

At six hours the pain moves to the right lower quadrant. The assistant has until eight hours to start antibiotics that cover gut bacteria. The scenario also has two complication events: if the patient is still untreated at 36 hours, the appendix perforates; and if the assistant reaches for a penicillin, some runs bring an allergic reaction.

The chart shows why timing matters: in the worst-case course, the risk of losing the crewmember is up to 8.7 percent if treated in time, and up to 50 percent if not. The scenario is scored on 19 checklist items plus the standard items for diagnosis, cause and each complication.

Plan of work: six phases, authoring to reports

Six phases; authoring is built, and the runner and scoring come next.

TALKING POINTS

- **A, scenario authoring: built.** Done when a clerk with no briefing builds a valid scenario.
- **A5–A7, next:** second clerk round; multi-patient and triage scenarios; a manual scoring sheet.
- **B, runner:** connect any assistant; one scenario end to end on the laptop overnight.
- **C, scoring:** scorer, judge and the anomaly viewer.
- **D, digital twin:** CLiFF course model, drug effects, complication events.
- **E, reports:** medical and technical, and checking the scorer against clerk scoring.
- B and C start on a **simple twin** so scoring can be tested early; D swaps in the full physiology.

NEXT “To keep that moving, we need four things.”

SCRIPT

Here's the plan of work. Phase A, scenario authoring, is built, and it's done when a clerk with no briefing can build a valid scenario. Next is a second round with clerks, scenarios with several patients and triage, and a sheet for scoring test runs by hand.

Phase B is the runner: connecting an assistant and running a scenario end to end on the laptop overnight. Phase C is scoring: the scorer, the judge and the viewer. Phase D is the full digital twin: condition courses from the CLiFFs, drug effects and complication events. Phase E is the reports, and checking the automated scorer against the clerks' own scoring.

B and C start on a simple twin, so we can test scoring early, and D swaps in the full physiology.

What we need to keep moving

Four asks; the Scenario Builder is ready now.

TALKING POINTS

- **Clerks:** a second round of scenarios on the Scenario Builder, timing each one.
- **Clinical lead:** review the rubric: critical errors, rating descriptions, dose tolerance.
- **Physiologist:** check the CLiFF course numbers and the crew baselines.
- **CMO-DA team:** share the interface spec before we build the runner.
- The **Scenario Builder is ready now**: one offline file, nothing to install.

NEXT “Thank you. Happy to take questions.”

SCRIPT

To keep this moving, we need four things. From the clerks, a second round of scenarios on the Scenario Builder, with the time each one takes. From our clinical lead, a review of the rubric: the critical errors, the rating descriptions and the dose tolerance. From our physiologist, a check of the CLiFF course numbers and the crew baselines. And from the CMO-DA team, the interface spec, before we build the runner.

The Scenario Builder is ready for clerks today: it's one file that opens offline, with nothing to install. Thank you. I'm happy to take questions.

Key numbers

Every figure in the deck, what it means and where it comes from.

Rubric figures are from the draft rubric, pending clinical-lead review. CLiFF figures were extracted automatically, pending physiologist review.

Figure	Meaning	Source
118	NASA IMM conditions (2023 CLiFF list)	NASA IMM documentation
17	Design Reference Missions	NASA IMM DRMs
8	medical kit and capability profiles (ISS, Orion, lunar lander, habitats)	IMM kit lists; gaps marked as assumptions
20	synthetic crewmembers	Modelled on the active corps (NASA cohort profile)
678	ICD-10-CM symptom codes	ICD-10-CM
90	reference entries: 73 kit drugs and 17 terms	FDA labeling (DailyMed); SpaceDoc glossary
100 / 89 / 108 / 89 / 106	conditions possible in LEO / cislunar / lunar surface / Mars transit / Mars surface	CLiFF incidence data
84 / 142	ISS drug / equipment item types	IMM kit
55 / 62	Orion drug / equipment item types	IMM optimized kit
35 / 23	lunar lander drug / equipment item types	IMM 7.5-day kit
Up to 12	runs per scenario: 5 environments × best and worst course, plus 2 at CLiFF odds	Design decision
~1 s to 23 min	one-way communication delay, LEO to Mars (3–23 min at Mars)	Environment settings
6 and 6	universal critical errors; global rating domains (1–5)	General rubric (draft)
−10 to +10	checklist item weights	General rubric (draft)
28	common checklist items clerks can add in one click	General rubric (draft)
+6 / +4 / +5 / +5	standard items: diagnosis, cause, complication recognised, complication response	General rubric (draft)
8.7% vs 50%	appendicitis worst case, maximum risk of loss of crew life: treated vs untreated	Appendicitis CLiFF, Table 4 (pending review)
62.6–77.5%	treated appendicitis cases that follow the best course	Appendicitis CLiFF, Table 4 (pending review)
T+8h / T+36h / 19	worked example: antibiotic deadline / perforation if untreated / checklist items	Worked example MSE-0001
~40 a year	clerks (medical students and residents) writing scenarios	Design doc
MacBook Air M4, 16 GB	where batches run to start, overnight	Design doc

Likely questions

Short answers you can give from the floor.

Why not just test with a question bank?

A question bank checks one answer at a time. It can't show whether an assistant re-checks the patient, handles a complication, or adapts to what's aboard. Here the simulated patient changes with the advice, and each scenario runs up to 12 times.

How do we know the automated scorer is right?

It scores one observable item at a time and has to cite the transcript turn or twin event. Every critical-error hit, and anything it can't decide, goes to a person. Clerks hand-score a calibration set, and the scorer is trusted only on items where it agrees with clerks about as well as clerks agree with each other (Phase E).

Does it work with CMO-DA?

Yes, by design. Any assistant connects through a small adapter for the conversation, and optionally through MCP tools to read vitals or order tests. We need the CMO-DA interface spec to confirm before building the runner.

Why run locally instead of in the cloud?

Nothing leaves the machine, there's no per-run API cost, and runs are reproducible. It starts on a laptop overnight and can move to bigger hardware later.

How long does a batch take?

Not measured yet. Phase B is done when one scenario runs end to end overnight on the laptop, which gives the first real timing.

Is the physiology real?

Each condition's course comes from NASA's CLiFFs: how often it goes well or badly, how long each phase lasts, and the risk of evacuation or loss of crew life. Vital signs follow that course, with any values the clerk sets. The numbers were extracted automatically and are pending physiologist review. A higher-fidelity engine (Pulse) is a later option.

Are these real astronauts?

No. They're 20 synthetic crewmembers, shaped to match the active corps (sex ratio, ages, backgrounds, training). Their histories are invented and pending physiologist review.

What about space weather, dust storms or communication blackouts?

Out of scope for now; the format already holds them as a later feature. Communication delay is included as a constant for each environment.

Is there one overall score?

No. The four layers are reported side by side, so one number never hides a critical error. A combined score is optional and never the headline.

Who writes the scenarios, and how long does one take?

Medical students and residents, about 40 a year, in the offline Scenario Builder. Time per scenario will be measured in the second clerk round.

When will we see results on SpaceDoc or CMO-DA?

After Phases B and C, the runner and scoring. No dates are set yet; the next steps are the second clerk round and the CMO-DA interface spec.

What's the biggest risk?

Trust in the automated scoring. That's why every score cites evidence, people confirm critical errors, and the scorer is checked against clerks before its numbers are used.